5.3.24 I²C interface characteristics

The I²C interface meets the timing requirements of the I2C-bus specification and user manual rev. 03 for:

- Standard mode (Sm): Bit rate up to 100 kbit/s.
- Fast mode (Fm): Bit rate up to 400 kbit/s.
- Fast mode plus (Fm+): Bit rate up to 1 Mbit/s.

The I²C timing requirements are specified by design, not tested in production, when the I²C peripheral is properly configured (refer to the product reference manual).

The SDA and SCL I/O requirements are met with the following restrictions:

- The SDA and SCL I/O pins are not true open-drain. When configured as open-drain, the PMOS connected between the I/O pin and V_{DDIOx} is disabled but remains present.
- Only FT_f I/O pins support Fm+ low level output current maximum requirement. Refer to Section 5.3.14: I/O port characteristics for the I²C I/Os characteristics.

Al I²C SDA and SCL I/Os embed an analog filter. Refer to the table below for the analog filter characteristics.

Table 64. I²C analog filter characteristics

Evaluated by characterization - Not tested in production.
Measurement points are taken at 50% V_{DD}.

Symbol	Parameter	Min	Max	Unit
t_{AF}	Maximum pulse width of spikes that are suppressed by analog filter	50(1)	160(2)	ns

1. *Spikes with widths below $t_{AF(min)}$ are filtered.*
2. *Spikes with widths above $t_{AF(max)}$ are not filtered.*

5.3.25 USART characteristics

Unless otherwise specified, the parameters given in Table 65 are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and V_{DD} supply voltage conditions summarized in not found, with the following configuration:

- Output speed set to OSPEEDRy[1:0] = 10
- Capacitive load C_L = 30 pF
- Measurement points done at 0.5 × V_{DD} level
- I/O compensation cell activated

Refer to I/O port characteristics for more details on the input/output alternate function characteristics (NSS, CK, TX, RX for USART).

Table 65. USART (SPI mode) characteristics

Evaluated by characterization - Not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CK}	USART clock frequency	Master transmitter mode, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	-	18	
		Slave receiver mode, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	-	48	
		Slave transmitter mode, 2.7 V ≤ V_{DD} ≤ 3.6 V	-	-	27	
$t_{su(NSS)}$	NSS setup time	Slave mode	t_{ker}(1) + 2	-	-	ns
$t_{h(NSS)}$	NSS hold time	Slave mode	2	-	-	
$t_{w(CKH)}$, $t_{w(CKL)}$	CK high and low time	Master mode	1/ f_{CK} /2-1	1/ f_{CK} /2	1/ f_{CK} /2+1	
$t_{su(RX)}$	Data input setup time	Master mode	18	-	-	